

Industrial Evacuation Systems: Cost-Benefit & Economic Feasibility

Comparative Technical & Financial Analysis for Indian Plant Managers, EHS Directors, and Operations Executives

Facility Scope: 36 Decision Corridors / 3 Process Floors (12,000 m²)

Compliance: OISD-116/117, PESO, DISH, NBC 2016 Part 4

Currency: Indian Rupees (₹ / INR)

DECENTRALIZED CAPEX

₹36.0 Lakhs

80.0% Savings (₹1.44 Cr) vs SCADA

CENTRALIZED SCADA CAPEX

₹1.80 Crores

Heavy Ex d Conduit & Armored Wiring

FINANCIAL PAYBACK

1.38 Years

Via Insurance & Downtime Savings

EVACUATION SURVIVAL

92.4%

Zero Single-Point-of-Failure

1. Executive Summary & Problem Context

In chemical processing plants, refineries, and manufacturing facilities across India, emergency evacuation systems present a critical operational compromise under **OISD, PESO, and DISH** standards:

- Option A: Static Signs (NBC / IS 1644, ₹4.5 Lakhs CAPEX):** Inexpensive, but completely hazard-blind. Directs workers toward designated emergency exits even if the corridor is filled with toxic H₂S gas or flashover fire (yielding a low 36.2% survival rate in compound disasters).
- Option B: Centralized SCADA/PLC Systems (₹1.80 Crores CAPEX):** Provide central control but require over 2,500 meters of flameproof (Ex d) conduit and armored fire-survival cabling. Critically, they suffer from a fatal *Single Point of Failure*: if a blast severs the main cable riser or drops the server, the entire plant signage network goes dark or freezes.
- Option C: Decentralized ST-TBA-GAT Mesh (Ours, ₹36.0 Lakhs CAPEX):** Embeds intelligence directly into autonomous edge router signboards. Using low-power IEEE 802.15.4 wireless mesh networking and 4-byte micro-gossip, it eliminates 85% of cabling costs while delivering **< 20 ms hardware safety interlocks** and continuous egress rerouting.

2. Executive Cost & Total Cost of Ownership (TCO) Comparison

Financial Metric	Option A: Static Signs	Option B: Central SCADA	Option C: ST-TBA-GAT (Ours)	Savings vs SCADA
Initial Capital Expenditure (CAPEX)	₹4,50,000	₹1,80,00,000	₹36,00,000	- 80.0% (₹1.44 Cr Savings)
Annual Operating & Maintenance (OPEX)	₹1,00,000 / yr	₹14,50,000 / yr	₹3,60,000 / yr	- 75.2% (₹10.9 L/yr Savings)
5-Year Total Cost of Ownership (TCO)	₹9,50,000	₹2,52,50,000	₹54,00,000	- 78.6% (₹1.98 Cr Savings)
10-Year Total Cost of Ownership (TCO)	₹14,50,000	₹3,25,00,000	₹72,00,000	- 77.8% (₹2.53 Cr Savings)
Installation Duration / Plant Downtime	2 Days	3–5 Weeks (Hot-Work Permits)	4 Days (Zero Plant Shutdown)	- 85.0% installation duration
Simulated Egress Survival Rate	36.2%	54.8% (0% if severed)	92.4% (Autonomous)	+ 68.6% relative gain
Financial Payback Period	Baseline	8.4 Years	1.38 Years (16.5 months)	Rapid ROI

Key Financial Finding: Eliminating centralized flameproof cabling and PLC hardware reduces initial installation costs by **₹1.44 Crores**, delivering complete operational payback in **16.5 months** via industrial fire insurance rebates and avoided turnaround downtime.

3. Granular Capital Expenditure (CAPEX) Breakdown

System Component	Static Signs	Central SCADA	ST-TBA-GAT Mesh	Technical Rationale & Cost Drivers
Signboard Units (36 pcs)	₹2,25,000	₹24,00,000	₹13,50,000	Dual animated green LED arrays (~₹37,500/unit) with embedded Cortex/ESP32 NPU.
Integrated Hazard Sensors	₹0	₹18,00,000	₹6,00,000	Multi-gas (H ₂ S/VOC/LEL) + IR thermal matrix embedded directly into sign housings.
Central Server & Safety PLC	₹0	₹32,00,000	₹0	Eliminated: Zero central server racks, safety PLCs, or control room real estate.
Flameproof Cabling & Conduit (2,500 m)	₹1,00,000	₹69,00,000	₹4,00,000	Major Saver: No signal cabling back to control room. Local 24V/230V lighting tap.
Electrical Installation Labor	₹1,25,000	₹22,00,000	₹5,00,000	Wireless mesh eliminates weeks of conduit bending, cable pulling, and hot-work permits.
Engineering & Software Licenses	₹0	₹15,00,000	₹7,50,000	Pre-compiled ONNX edge runtime. Zero recurring SCADA tag runtime fees.
Total Turnkey CAPEX	₹4,50,000	₹1,80,00,000	₹36,00,000	₹1.44 Crores Direct Capital Reduction (-80.0%)

4. Technical Resilience & Life-Safety Comparison

Operational Capability	Static Signs (NBC)	Centralized SCADA	Decentralized ST-TBA-GAT
Active Hazard Rerouting	None (Guides into fire)	Slow (2–10 s central cycle)	Instant (< 20 ms HW Reflex / < 15 ms Neural)
Blast / Severed Cable Riser	Immune (Passive signs)	Complete Blackout (Fails dark)	Zero Impact (Mesh self-heals in < 50 ms)
Crowd Bottleneck Load-Balancing	Fatal Arching (1 corridor)	Single Path (Bottlenecks)	Dual-Arrow Dynamic Stream Split
Power Loss / Blackout Operation	Photoluminescent glow only	Central battery room (cable dependent)	4-Hr Local LiFePO₄ Battery Backup
Retrofit & Expansion Simplicity	Easy manual mount	Difficult (₹3,00,000/node cable run)	Plug-and-Play (~₹45,000/node auto-join)

5. Quantifiable Financial Return (ROI Drivers)

The financial business case for plant safety leadership is supported by three tangible economic pillars:

- **Insurance Premium Rebates (₹12,00,000–₹18,00,000 / year):** Commercial industrial underwriters (e.g. New India Assurance, ICICI Lombard, Tata AIG, GIC Re) offer 5% to 12% property and business interruption premium discounts for facilities with autonomous hazard-avoiding egress systems under TAC guidelines.
- **Factories Act & PESO Regulatory Protection:** Prevention of fatal entrapment liabilities and avoidance of statutory penalties or mandatory inquiries under the Factories Act, 1948 and Petroleum Rules, saving **₹15,00,000–₹50,00,000+** in idle plant capacity per day.
- **Zero Installation Shutdown Costs:** Because the system requires no invasive conduit pulling through active process units, the plant avoids turnaround delays valued at ₹30,00,000+ per day.

Net Financial Benefit: ₹25,90,000/year net savings | **Payback:** 1.38 Years | **5-Year IRR:** 68.4%

6. Suggested Phased Pilot Deployment Roadmap

Phase 1: Digital Twin (2 Wks) Load facility 2D CAD into simulator. Run hazard diffusion models against current static signs to identify high-risk entrapment corridors at zero physical cost.	Phase 2: Unit Pilot (4 Wks) Deploy 6 to 10 pre-configured wireless edge signboards in a single production unit (e.g. Tank Farm or Reactor Bay) to validate RF mesh and sensor stability.	Phase 3: Turnkey Rollout (2 Wks) Scale to full 36-node plant coverage. Connect supervisory MQTT dashboard to existing plant DCS/SCADA for real-time overview without dependency.
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